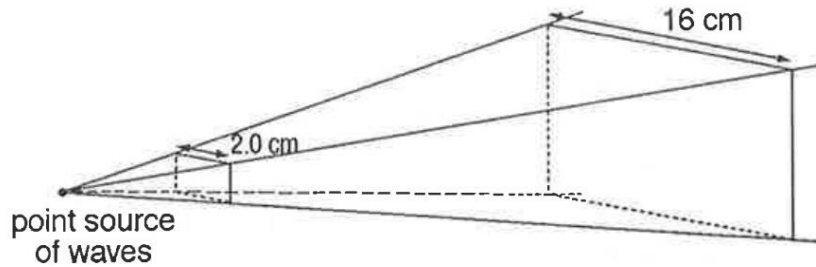


- 15 Waves from a point source pass through an area that is 2.0 cm wide as shown below.



Within this area, the intensity of the waves is I and their amplitude is A . The waves reach a second area of width 16 cm.

What will be the intensity and amplitude of the waves when they reach the second area?

	Intensity	amplitude
A	$\frac{I}{8}$	$\frac{A}{4}$
B	$\frac{I}{64}$	$\frac{A}{4}$
C	$\frac{I}{64}$	$\frac{A}{8}$
D	$\frac{I}{256}$	$\frac{A}{16}$

Ans: **C**

$$I \propto \frac{A^2}{\text{area}}$$

$$I(2 \times 2) = I_{\text{new}}(16 \times 16)$$

$$I_{\text{new}} = \frac{I}{64}$$

$$\frac{I}{A^2} = \frac{I_{\text{new}}}{A_{\text{new}}^2}$$

$$A_{\text{new}} = \frac{A}{8}$$

- 16 Which of the following is a necessary condition for clear and stable interference patterns to be